FEDERAL SPECIFICATION

TEST TUBES, BLOOD SEDIMENTATION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers blood sedimentation tubes suitable for determining sedimentation rates and the volume of packed red blood cells.

1.2 Classification.

1.2.1 *Types and sizes.*—Sedimentation tubes covered by this specification shall be of the following types and sizes, as specified:

Type I.—Cutler
Size.—1 ml.
Size.—5 ml.
Type II.—Wintrobe Hematocrit

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

DD–G–541—Glass (Laboratory).
DD–G–626—Glassware, Laboratory; Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.—Tubes shall be fabricated of glass in strict conformance to type I or type II of Federal Specification DD–G–541.

3.2 Style and design.

3.2.1 *Type I (Cutler).*—The tube shall have a cylindrical body terminating in a round bottom. The wall of the open end shall be cut at right angles to its longitudinal axis and finished smoothly without lip. Tubes shall be constructed in strict accordance with figure 1 for size specified.

3.2.1.1 *Graduated scale.*—Each tube shall be graduated in millimeters, as shown in figure 1, for size specified. The scale shall extend downward in 1 mm. divisions to 40 mm. The zero (0) graduation at the top, and each tenth graduation mark downward, shall be numbered. There shall also be a graduation mark without number as shown in figure. Each 5 mm. graduation mark shall be at least one-third longer than the 1 mm. markings. The position of the numerals shall remain consistent throughout the length of the graduated scale. All markings shall be sharp and distinct. The graduation markings shall be at right angles with the long axis of the tube.

3.2.2 *Type II (Wintrobe Hematocrit).*—The tube shall have a cylindrical body terminating in a round bottom on the outside and a perfectly flat bottom on the inside. The tube shall have a uniform bore and a beaded flared top. Tubes shall be in strict accordance with figure 1. Each tube shall be supplied with a rubber cap.

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3.2.2.1 *Graduated scale.*—Each tube shall be graduated in millimeters as shown in figure 1. One set of graduations shall be from 0 to 10 cm. in 1 mm. divisions, and the other from 10 to 0 cm. in 1 mm. divisions. Scale graduations shall be accurate to ±0.25 mm. when measured from the middle of the inside bottom of the tube. The two sets of numerals shall be either in contrasting colors (red and blue or white), or in one color. The graduation markings shall be at right angles with the long axis of the tube. All markings shall be sharp and distinct.

3.3 Marking.

3.3.1 *Marking methods.*—Inscriptions, graduation markings, and numbers shall be either etched and filled with permanent pigment, or a pigment fired into glass (not ceramically imprinted).

3.3.2 *Permanency of markings.* — The appearance of the lines, numbers, and inscriptions shall be the same, as judged by eye, before and after each test described in 4.3.1.

3.4 Identification marking.—Tubes shall be permanently and legibly marked with the name or trade-mark of the manufacturer. In addition, there shall be a sandblasted area for marking purposes.

3.5 Workmanship.—Workmanship shall be first class throughout. Tubes shall be free from defects which detract from their appearance or which may impair their serviceability. All apparatus shall be well annealed so that when tested as specified in 4.3.2, there shall be no ring strain. Longitudinal strain, if present, shall be faint and highly diffused.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. — Unless otherwise specified, at least six tubes shall be taken at random from each lot of 100, or fraction thereof delivered, for inspection and tests.

4.2 Inspection.—Samples shall be inspected for materials, style and design, identification marking, workmanship, packaging, packing, and marking.

4.3 Tests.

4.3.1 *Permanent color of markings.*—In performing the following, a different sample shall be used for each test.

4.3.1.1 *Alkali test.*—Immerse samples completely in boiling normal sodium hydroxide, cover, and boil 1½ hours. During boiling period make up evaporative losses with boiling distilled water. Do not permit sodium hydroxide to concentrate itself more than 5 percent over the initial strength. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.3.1.2 *Acid test (hydrochloric).* — Immerse samples completely in boiling concentrated hydrochloric acid (sp. gr. 1.18) and let boil for 1½ hours, replacing loss by fresh concentrated acid. During boiling, do not let the volume decrease more than 5 percent before adding fresh acid. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.3.1.3 *Acid test (sulphuric).*—Immerse samples completely in concentrated sulphuric acid at room temperature. Let stand 1 hour. Remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.3.2 *Strain examination.* — Tubes shall be tested in accordance with method No. 420 of Federal Specification DD–G–626 to determine requirements for strain.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, tube shall be packaged in accordance with accepted commercial practice.

5.2 Packing.—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages.*—Unless otherwise specified, packages shall be marked with the name and size of the commodity, the quantity contained therein, and the name of the manufacturer and contractor.

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5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name and size of the commodity as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Ordering data.—Purchasers should specify type and size required, and should exercise any desired options offered herein. (See 1.2.1, 4.1, 5.3, 6.3, and 7.2.3.)

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

6640–445–1700	Test Tube, Blood Sedimentation, Cutler, 1 ml.
6640–445–6100	Test Tube, Hermatocrit Determination, Wintrobe.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine conformance to the requirements of this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent notice.—The furnishing by the Government of specifications, including drawings or other data in connection therewith, does not in any way grant the recipient of such material, or any other person, a license or right to manufacture, use, or sell any patented or unpatented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN–B–621—Boxes, Wood, Nailed and Lock-Corner.
LLL–B–631—Boxes; Fiber Corrugated (for Domestic Shipment).
PPP–B–676—Boxes, Set-Up, Paperboard.

Military Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
MIL–L–10547—Liners, Case, Waterproof.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

7.2.2 *Packaging.*

7.2.2.1 *Unit of issue.*—One tube, as specified, constitutes one unit.

7.2.2.2 *Unit packaging.*

7.2.2.2.1 *Type I tubes.*—Unit packaging is not required.

7.2.2.2.2 *Type II tubes.*—Each unit shall be wrapped in tissue.

7.2.2.3 *Intermediate packaging.*

7.2.2.3.1 *Type I tubes.*—Twelve units shall be packaged in a container of suitable size and design. Container shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Twelve containers (144 units) shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification LLL–B–631 or PPP–B–676. Each container and each carton shall be adequately secured to prevent accidental opening.

7.2.2.3.2 *Type II tubes.*—Twelve wrapped tubes (12 units) shall be packaged in a carton of appropriate size constructed in accordance

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with Federal Specification LLL–B–631 or PPP–B–676. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.*—The number of intermediate cartons specified in table I shall be overpacked in an exterior container designed for type 3 load. Construction, closure, and strapping (when required) shall be in accordance with the requirements of Federal Specification NN–B–621 or LLL–B–631.

7.2.3.2 *For overseas shipment.*—The number of intermediate cartons specified in table I shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with the requirements of Military Specification JAN–P–106.

TABLE I.—*Packing quantities*

Stock No.	Type	Number of intermediate cartons to be overpacked in each shipping container	Total number of units to be contained in each shipping container
6640–445–1700	I	12	1,728
6640–445–6100	II	36	432

7.2.3.2.1 *Case liner.* — Each overseas box shall be lined with a waterproof case liner conforming to the requirements of Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

7.2.3.2.2 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

NOTE.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

7.2.3.3 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

7.2.4 *Marking.*—Each intermediate container and carton, and each shipping container shall be marked in accordance with Military Standard MIL–STD–129.

7.2.5 *Ordering data.* — Procurement documents should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.3.1 and 7.2.3.2.)

MILITARY INTERESTS:

Army—M
Navy—Md
Air Force.

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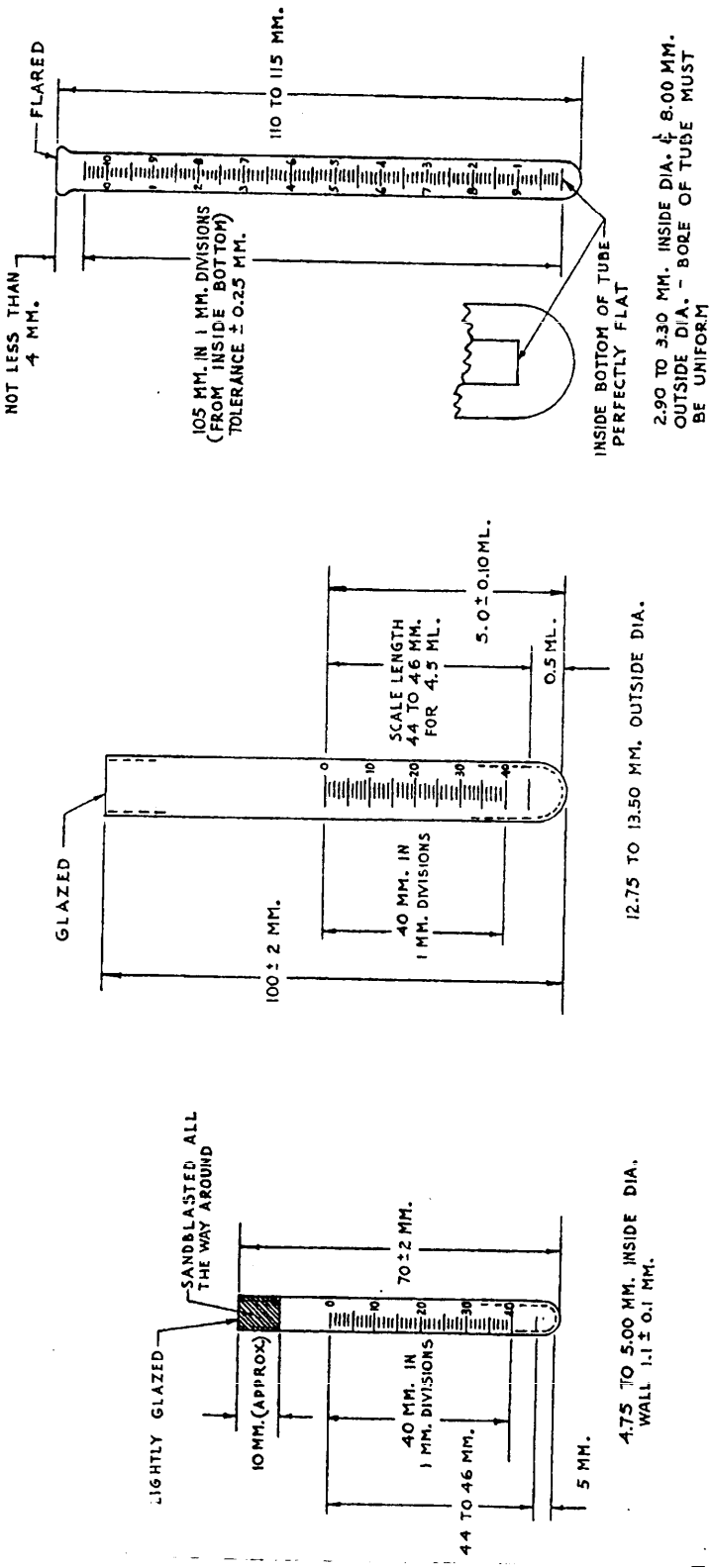


FIGURE 1

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